

LOGISTICS OPERATIONS • 3PL GOVERNANCE •
ANALYTICS

Supply Chain Operations & Logistics Performance Improvement

Establishing KPI governance, provider accountability and exception control across a multi-site distribution network

72%

BASELINE ON-TIME

1,248

SHIPMENTS
ANALYSED

11%

EXCEPTION RATE

46%

CAUSES ARE
INTERNAL

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Portfolio disclosure. This is a simulated portfolio project using synthetic data. Apex Distribution Ltd. is fictional. All operating figures are constructed for demonstration and do not represent a client engagement or confidential employer information.

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SECTION 1

Executive Summary

The problem

Apex Distribution Ltd. operates eight stocking and distribution locations, six external logistics providers and 850 active SKUs. Across a synthetic 12-week dataset of **1,248 shipments**, on-time delivery sits at **72%** against a 95% service expectation, with an exception rate of **11%**. Each provider reports in a different format, so management cannot reliably separate carrier failures from internal process failures.

What was done

I designed a logistics performance management system rather than a reporting layer. That meant standardising the shipment data structure and exception reason codes, defining one KPI dictionary applied identically across providers and sites, building a weighted 3PL scorecard, designing a detect-to-close exception workflow with named owners, and connecting all of it to a weekly and monthly governance cadence.

The finding that changes the conversation

When the 137 exceptions were classified by ownership, **46% traced to internal process failures**, principally late shipment release and missed provider cut-offs, against 28% attributable to the logistics providers. The performance problem is substantially internal, and the operating model makes that visible for the first time.



The headline insight

Late delivery is an outcome, not a cause. Almost half of all exceptions originate inside the business, before the carrier ever collects the shipment. Any improvement programme that begins by pressuring providers would address barely a quarter of the problem.

Method note. All figures derive from a synthetic 1,248-shipment dataset constructed to match the operating profile described in this case study. The dataset, KPI model, scorecard and dashboards are supplied as a companion Excel workbook.

SECTION 2

Business Context

Apex Distribution Ltd. is a fictional regional distribution company supplying customers from multiple stocking locations across a regional network. The business has grown faster than its logistics reporting and governance processes.

Operating metric	Value
Stocking and distribution locations	8
External logistics providers	6
Active SKUs	850
Shipments analysed	1,248
Analysis period	12 weeks
Baseline on-time delivery	72%
Exception rate	11%
Total logistics spend in period	£100,812

Each logistics provider submits information in a different format. Operational teams rely heavily on spreadsheets and email, and performance reviews tend to focus on individual incidents rather than recurring patterns.

Four operational problems

- **Limited visibility** across logistics providers, with no common view of performance.
- **Inconsistent measurement**, as the same KPI is calculated differently by different parties.
- **Difficulty identifying root causes**, so symptoms are treated instead of drivers.
- **Weak accountability** for corrective actions, with no named owner or due date.

Core finding from the assessment

The central issue is not simply late delivery. It is the absence of a common operating model for logistics performance management. Without shared definitions and ownership, every performance conversation becomes a negotiation about whose numbers are correct.

SECTION 3

Business Requirements

I structured the improvement programme around six practical requirements, each written so that it can be tested after implementation.

#	Requirement	Test of success
1	Create one definition for each core logistics KPI	A published KPI dictionary used by all providers
2	Make provider performance comparable across sites and service types	Single scorecard ranks all six providers on common criteria
3	Identify the true owner of each exception	Every exception carries a root-cause category and named owner
4	Reduce time to identify and close operational issues	Exception closure time measured and trending down
5	Link service performance with shipment cost	Cost-to-service visible by provider, site and service type
6	Establish a repeatable governance process	Weekly operational review and monthly provider review in place

The governing principle

Performance measurement must lead to operational action. A KPI that nobody owns, reviews, or acts upon is an expense, not a control.

SECTION 4

My Role & Consulting Approach

In this simulated engagement I act as the supply chain operations consultant accountable for the full arc, from current-state assessment through to governance design.

Scope of my responsibility

- Current-state assessment
- Process mapping
- KPI definition and dictionary
- Logistics performance analysis
- 3PL and vendor performance management
- Exception management design
- Reporting and dashboard requirements
- Implementation planning and governance design

Capabilities demonstrated

Area	Capability
Operations	Process mapping, control design
Vendor	3PL scorecard, supplier review
Analytics	KPI modelling, Excel/BI dashboards
Governance	Exception workflow, escalation

Five principles I applied

Principle	What it means in practice
1. Process first, reporting second	A dashboard cannot fix an undefined process. The operating model and data definitions come first.
2. Measure the same thing the same way	Different providers must not use different definitions for the same KPI.
3. Separate symptoms from root causes	A late shipment is an outcome. The investigation must establish why it became late.
4. Assign every material exception an owner	Visibility without ownership does not create improvement.
5. Use visibility to drive decisions	Reporting exists to identify where management intervention is required.

SECTION 5

Current-State Assessment

The assessment identified five control gaps. Each was subsequently confirmed in the shipment data.

Control gap	Observed condition	Evidence in the data
KPI inconsistency	Providers and locations use different definitions and formats	No comparable baseline existed before the KPI dictionary was applied
Limited exception visibility	No consistent method to identify, categorise, assign and track exceptions	137 exceptions had no owner or reason code before standardisation
Manual reporting	Performance consolidated by hand from multiple sources	Six provider formats reconciled manually each week
Weak documentation	Timestamps, POD and references incomplete	POD completeness 94.0%, below the 99% control threshold
Inconsistent governance	Recurring issues handled as isolated incidents	42 exceptions still open or pending at period end

Cut-off compliance, the hidden driver

The assessment surfaced one condition that no provider report would ever reveal: across the network, roughly **half of all shipments are released after the provider daily cut-off time**. Because this happens inside the business, it is invisible in carrier performance data yet it directly causes downstream delay.

Why this matters

A shipment released after cut-off cannot be collected the same day. The delay is booked against the provider, the provider disputes it, and the review becomes an argument rather than a corrective action. This single blind spot explains a large share of the 28-point gap to target.

SECTION 6

Solution Architecture

I designed the operating model as five connected layers. Each layer is only useful if the one beneath it is in place.

Layer	Purpose	Key content
1. Standardised data	One shipment structure for every provider	Shipment ID, provider, origin, destination, service type, release, pickup, promised and actual delivery, cost, exception code, owner, closure status
2. Performance	One KPI set calculated identically	Eight KPIs spanning service, control, responsiveness and cost
3. Exception management	A consistent lifecycle for every issue	Detect, classify, assign, escalate, correct, verify, close
4. Governance	A cadence that forces follow-through	Weekly operational review, monthly provider scorecard, corrective-action tracking
5. Reporting	Drill-down from symptom to action	Four dashboard views with a single navigation path

The exception lifecycle

Detect → Classify → Assign → Escalate → Correct → Verify → Close

The reporting drill-down path

KPI → Provider / Site → Shipment → Exception → Root Cause → Owner → Corrective Action

Why this is a closed loop

Most logistics reporting stops at the KPI. This architecture continues through to a named owner and a verified corrective action, which is what converts a reporting solution into a management system.

SECTION 7

Logistics KPI Framework

Eight KPIs, each with a single definition, a stated purpose, a target and a measured baseline. This table is the published KPI dictionary.

KPI	Definition	Purpose	Target	Baseline
On-Time Delivery	Shipments delivered within the agreed window ÷ total delivered	Customer-facing reliability	95%	72.0%
Pickup Adherence	Shipments collected within the agreed pickup window ÷ total	Provider execution at origin	95%	82.9%
Transit Variance	Actual transit days minus expected transit days	Route and provider performance	0.0 days	-0.47 days
POD Completeness	Deliveries with valid proof of delivery ÷ total delivered	Documentation and invoice validation	99%	94.0%
Exception Rate	Shipments with one or more exceptions ÷ total	Operational instability	<5%	11.0%
Exception Closure Time	Days from exception creation to verified closure	Responsiveness and discipline	3.0 days	5.4 days
Cost per Shipment	Total shipment cost ÷ shipment count	Cost efficiency	£75.00	£80.78
Provider Reliability	Weighted composite of service, control, cost, communication and improvement	Supplier governance	80	50.7

Reading the baseline

Every KPI sits below target, but they fail differently. Pickup adherence at 82.9% points to origin-stage execution, POD at 94.0% points to documentation control, and closure time at 5.4 days points to management responsiveness. Separating these prevents a single blunt conversation about “service”.

SECTION 8

Performance Analysis & Findings

The standardised dataset made the first like-for-like comparison possible across twelve weeks, six providers and eight sites.

8.1 Network trend

On-time delivery oscillates between 65% and 80% with no improving trend, and exception rate moves inversely, confirming the two are linked. The absence of a trend is itself the finding: without a governance cadence, performance drifts rather than improves.

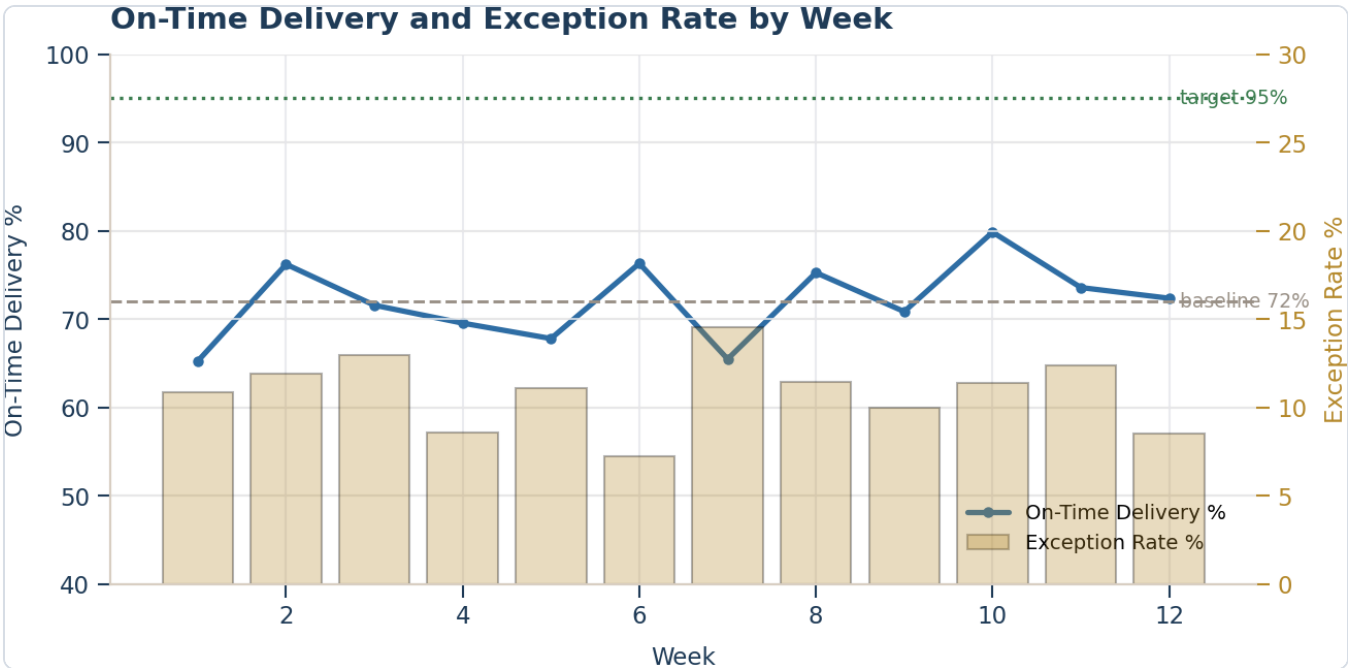


Figure 8.1 — On-time delivery and exception rate by week against the 72% baseline and 95% target.

8.2 Provider performance

Provider on-time performance spans a **34-point range**, from 82.4% to 48.0%. That spread is the single strongest argument for a common scorecard: the network average of 72% conceals both an acceptable performer and a failing one.

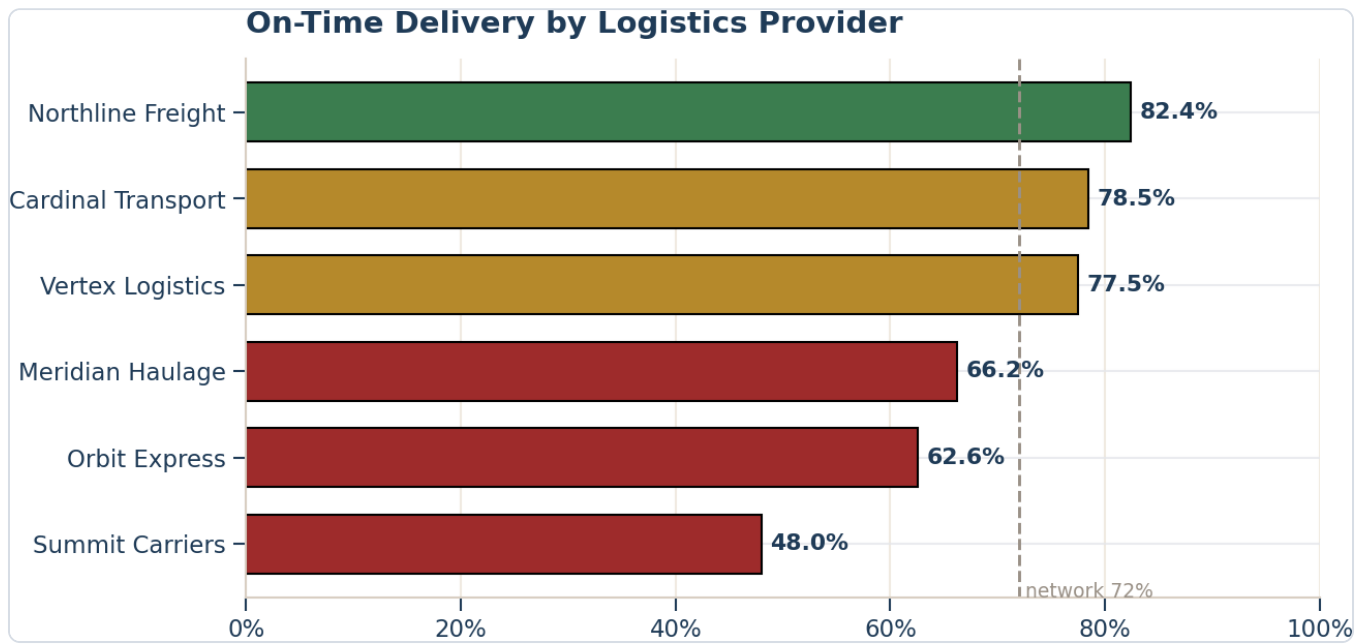


Figure 8.2 — On-time delivery by provider. Summit Carriers at 48.0% sits 24 points below the network average.

8.3 Site performance and cut-off compliance

Site on-time performance ranges from 77.9% to 65.4%. Critically, cut-off breach rates sit near 50% at every site, which shows this is a systemic process failure rather than a local one. Southampton combines the lowest on-time rate with an elevated exception rate, marking it as the first site for intervention.

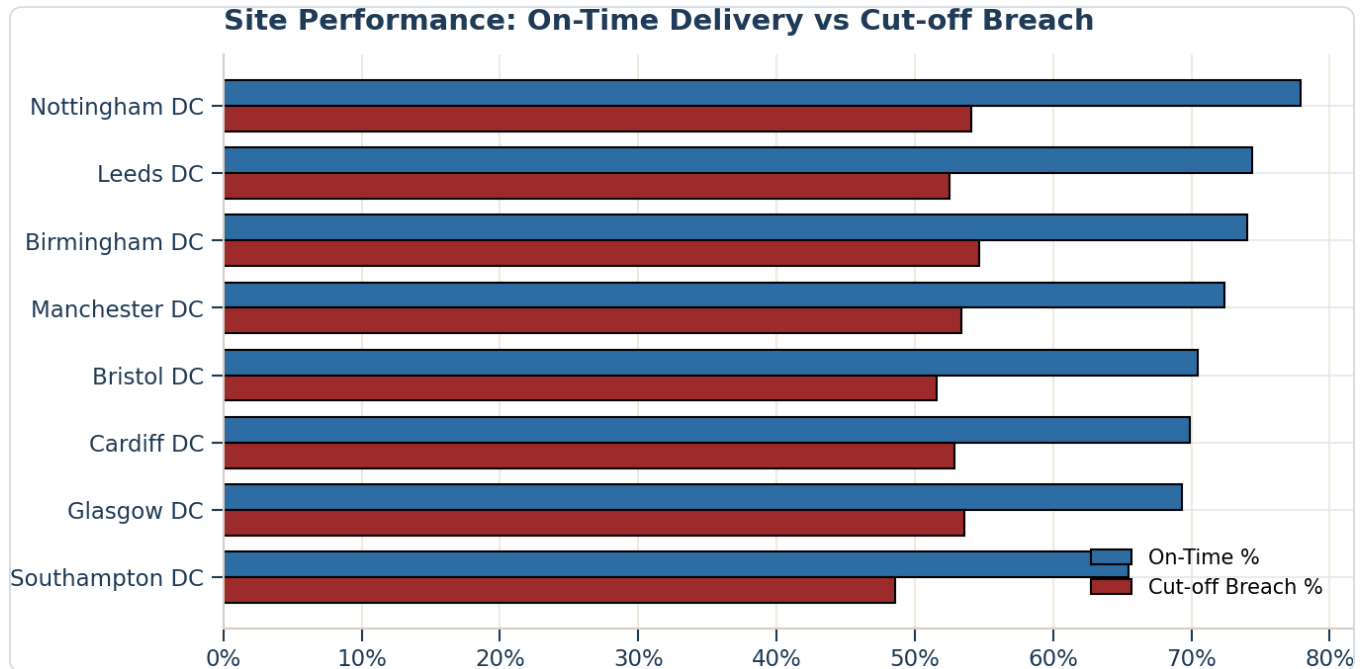


Figure 8.3 — Site on-time delivery against cut-off breach rate. The breach rate is uniformly high, indicating a network-wide release discipline problem.

8.4 Cost-to-service relationship

Linking cost with service exposes a misalignment that neither metric reveals alone.

Service type	Shipments	On-Time %	Cost per shipment	Observation
Next Day	239	66.1%	£119.00	Highest cost, worst reliability
Express	275	69.5%	£92.10	Premium price, below-average service
Standard	477	74.2%	£52.00	Core volume, above network average
Economy	166	75.9%	£36.70	Cheapest and more reliable than premium
Dedicated	91	76.9%	£177.50	Best service, highest unit cost

The cost-to-service contradiction

The business pays a **£82 premium** per Next Day shipment over Economy and receives **10 percentage points worse** on-time performance. Customers are being charged for speed the network cannot deliver. This is a commercial exposure as much as an operational one, and it is only visible once cost and service sit in the same model.

Root Cause Analysis

Every exception was classified into one of four ownership categories. This classification is what prevents each late shipment from automatically becoming a carrier-performance issue.

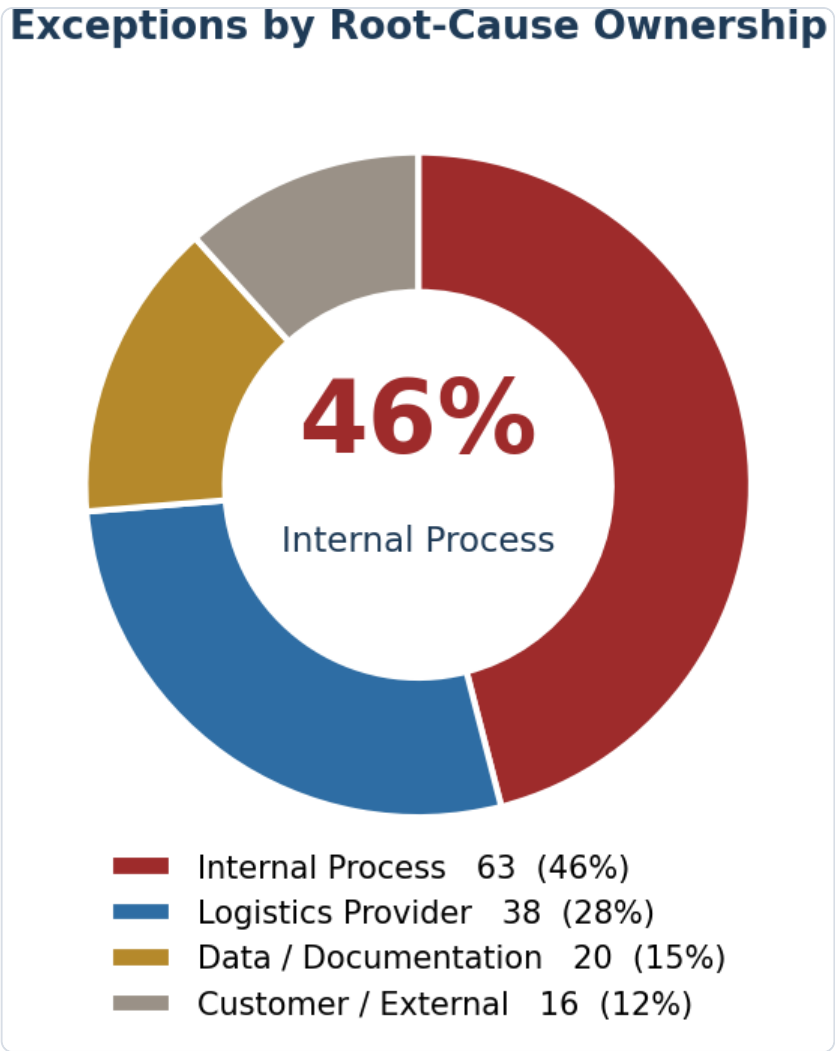


Figure 9.1 — Exceptions by root-cause ownership. Internal process failures are the largest single category.

Ownership category	Exceptions	Share	Representative causes
Internal Process	63	46%	Late shipment release, missed cut-off, incorrect booking details
Logistics Provider	38	28%	Transit delay, pickup failure, capacity constraint
Data / Documentation	20	15%	Missing POD, incorrect timestamps, missing references
Customer / External	16	12%	Customer unavailable, access restriction, external disruption

The conclusion this forces

If the business had opened this programme by challenging its providers, it would have been addressing 28% of the problem while the largest 46% continued untouched. Causality must be established before accountability is assigned.

9.1 Exception detail by reason code

At code level the picture sharpens further. The two largest reason codes are both internal, and together they account for **56 of 137 exceptions, 41% of the total**.

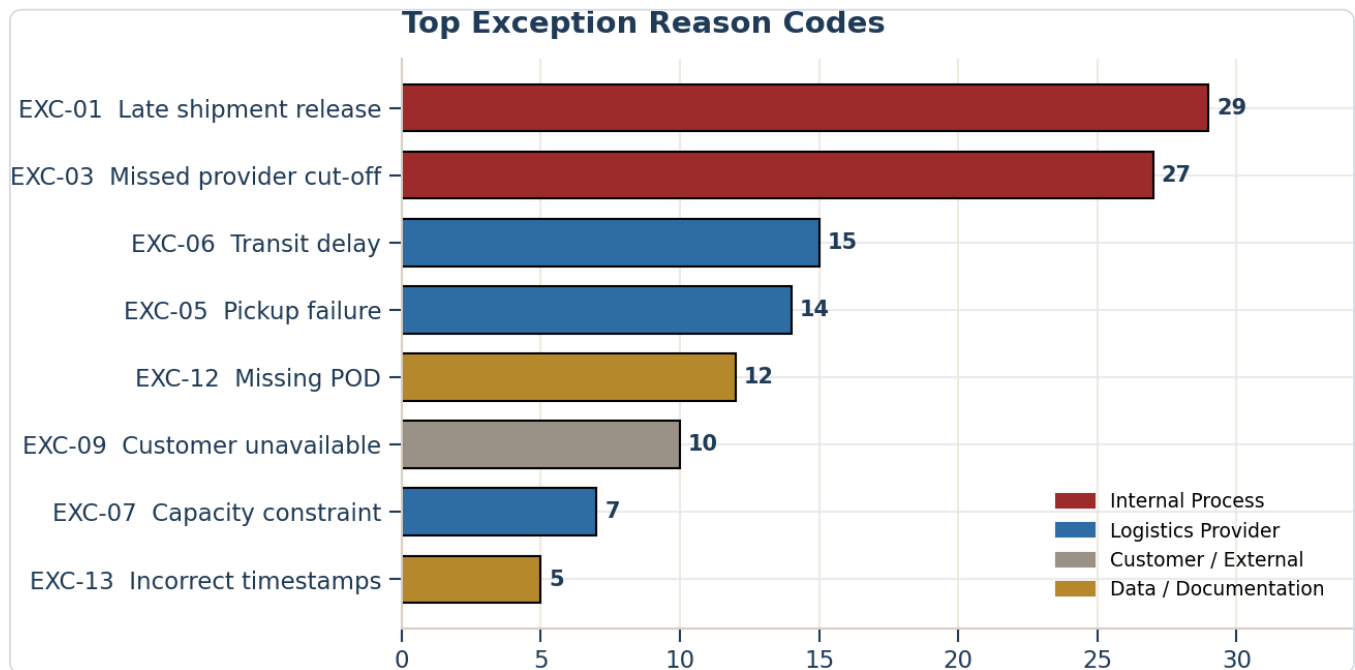


Figure 9.2 — Top exception reason codes, coloured by ownership category.

Code	Description	Ownership	Owner	Count	Avg closure
EXC-01	Late shipment release	Internal Process	Operations	29	7.3 days
EXC-03	Missed provider cut-off	Internal Process	Operations	27	4.4 days
EXC-06	Transit delay	Logistics Provider	Provider	15	3.6 days
EXC-05	Pickup failure	Logistics Provider	Provider	14	7.2 days
EXC-12	Missing POD	Data / Documentation	Logistics Admin	12	7.2 days
EXC-09	Customer unavailable	Customer / External	Customer Service	10	5.4 days

9.2 Ownership and responsiveness

Assigning exceptions to a named owner also exposes where closure discipline is weakest.

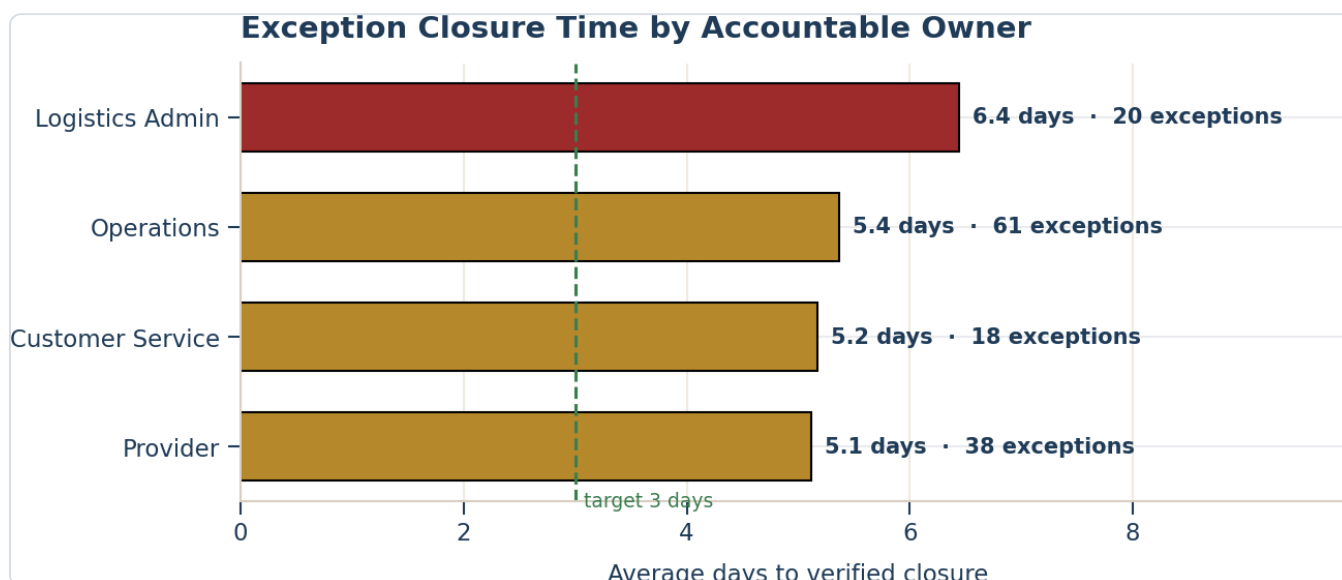


Figure 9.3 — Average exception closure time by accountable owner against the 3-day target.

Owner	Exceptions	Avg closure	Open / pending
Operations	61	5.4 days	24
Provider	38	5.1 days	9
Logistics Admin	20	6.4 days	4
Customer Service	18	5.2 days	5

No owner meets the 3-day target, and Operations carries both the largest volume and the largest open backlog. That is a capacity and prioritisation issue, not an attitude issue, and it should be addressed as such in the weekly review.

SECTION 10

3PL Provider Scorecard

I built a weighted scorecard so that supplier reviews rest on common criteria rather than on the most recent incident.

Dimension	Weight	What it captures
Service Reliability	35%	On-time delivery performance
Operational Control	25%	Pickup adherence and POD completeness
Cost Performance	20%	Cost per shipment relative to peers
Communication	10%	Exception closure responsiveness
Continuous Improvement	10%	Exception rate trend

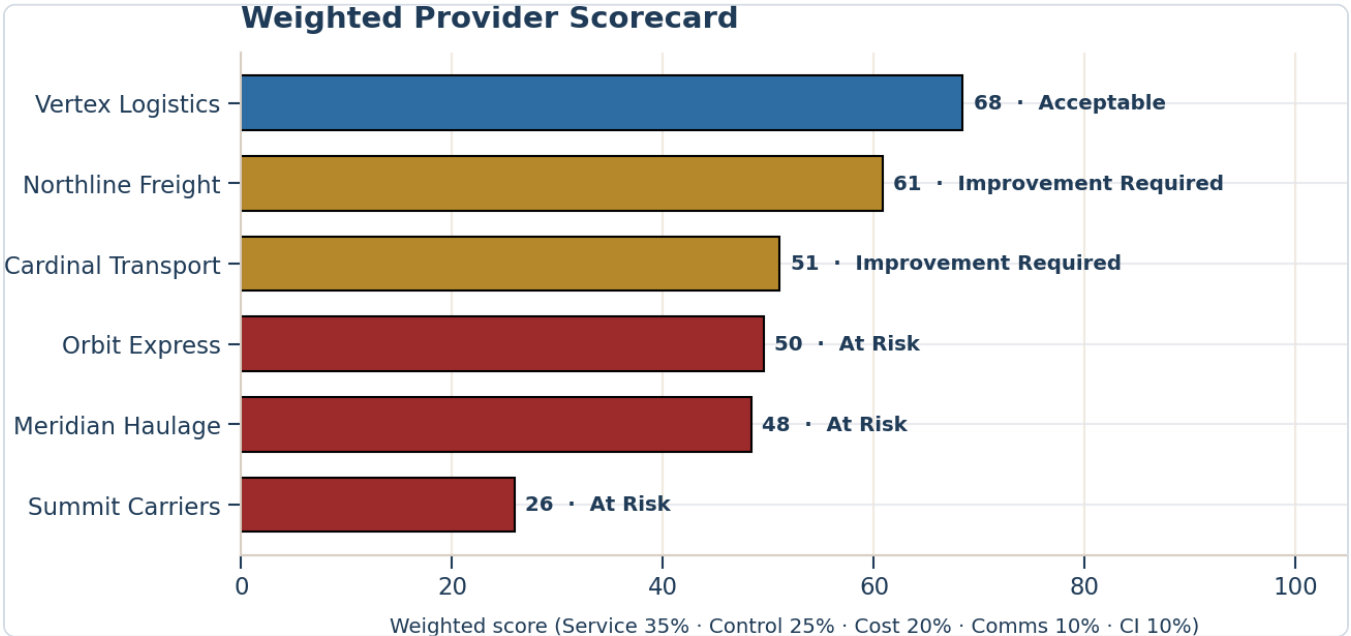


Figure 10.1 — Weighted provider scorecard with performance banding.

Provider	Ships	On-Time	Pickup	POD	Exc %	Cost	Score	Rating
Vertex Logistics	249	77.5%	83.1%	91.6%	10.0%	£78.20	68.4	Acceptable
Northline Freight	295	82.4%	82.7%	95.9%	8.5%	£81.50	60.8	Improvement
Cardinal Transport	209	78.5%	82.3%	94.7%	10.0%	£82.00	51.1	Improvement
Orbit Express	155	62.6%	81.9%	94.2%	14.2%	£80.50	49.6	At Risk
Meridian Haulage	213	66.2%	83.1%	93.0%	11.3%	£80.90	48.4	At Risk
Summit Carriers	127	48.0%	84.3%	94.5%	15.7%	£82.30	25.9	At Risk

10.1 What the scorecard reveals that a single KPI cannot

Best service is not the same as best value

Northline Freight has the highest on-time delivery at 82.4%, yet Vertex Logistics ranks first overall. Vertex delivers comparable reliability at a lower cost per shipment with a similar exception profile. Ranking providers on on-time delivery alone would have produced the wrong preferred supplier.

Summit Carriers requires immediate action

At 48.0% on-time and a 15.7% exception rate, Summit fails on both service and stability, scoring 25.9 against a network average of 50.7. Interestingly its pickup adherence is the highest in the network at 84.3%, so the failure occurs in transit, not at collection. That distinction directs the corrective conversation precisely.

10.2 How the scorecard is used

- Identify performance trends across periods rather than isolated incidents.
- Trigger corrective actions when a provider drops a performance band.
- Support structured monthly supplier reviews with common evidence.
- Identify recurring provider weaknesses by dimension, not by anecdote.
- Inform contract and volume allocation discussions.

Note on weights. The weights are illustrative. In a live engagement I would validate them with operations, procurement, finance and commercial stakeholders before implementation, because the weighting encodes what the business values.

SECTION 11

Exception & Escalation Process

Every material exception follows one workflow, with a named owner and closure evidence.

Stage	Action	Control
Detect	Identify a shipment that has breached or is likely to breach an agreed condition	Automated threshold on promised delivery
Classify	Assign the correct exception category and reason code	Standard reason-code list, 14 codes
Assign	Identify the accountable owner	Owner derived from root-cause category
Escalate	Escalate by severity, customer impact, age or service threshold	Defined escalation thresholds
Correct	Implement the immediate corrective action	Action recorded against the exception
Verify	Confirm the corrective action resolved the issue	Verification before closure
Close	Record evidence and formally close	Closure evidence mandatory

Worked example: the cut-off failure

A shipment misses its promised delivery window. The initial assumption is that the logistics provider caused the delay. Investigation shows the shipment was released after the provider's daily cut-off time. The root cause is therefore internal, not provider-related.

Late delivery → Assumed carrier fault → Investigation → Released after cut-off → Internal ownership

Corrective actions

- Cut-off compliance alerts before the daily deadline.
- Daily late-release reporting by site.
- Named internal ownership for late releases.
- Weekly review of recurring late-release patterns.

Why this example matters

EXC-01 and EXC-03 together represent 41% of all exceptions in the dataset. This single worked example, applied systematically, addresses the largest failure mode in the network.

SECTION 12

Dashboard & Reporting Design

Four dashboard views, built in Excel and supplied as a companion workbook, each answering a different management question. The design supports drill-down from a KPI variance to the underlying corrective action.

View	Question it answers	Key content
Executive Overview	Is the network performing, and is it improving?	On-time delivery, exception rate, cost per shipment, open exceptions, weekly trend
Provider Performance	Which providers need intervention?	On-time, pickup adherence, POD, exception rate, cost, weighted score and rating
Operations Exceptions	What is open, who owns it, and how old is it?	Open vs closed, aging, root-cause category, owner, closure time
Site & Route	Where in the network is performance lost?	Performance by origin, cut-off breach, transit variance, cost patterns, service type

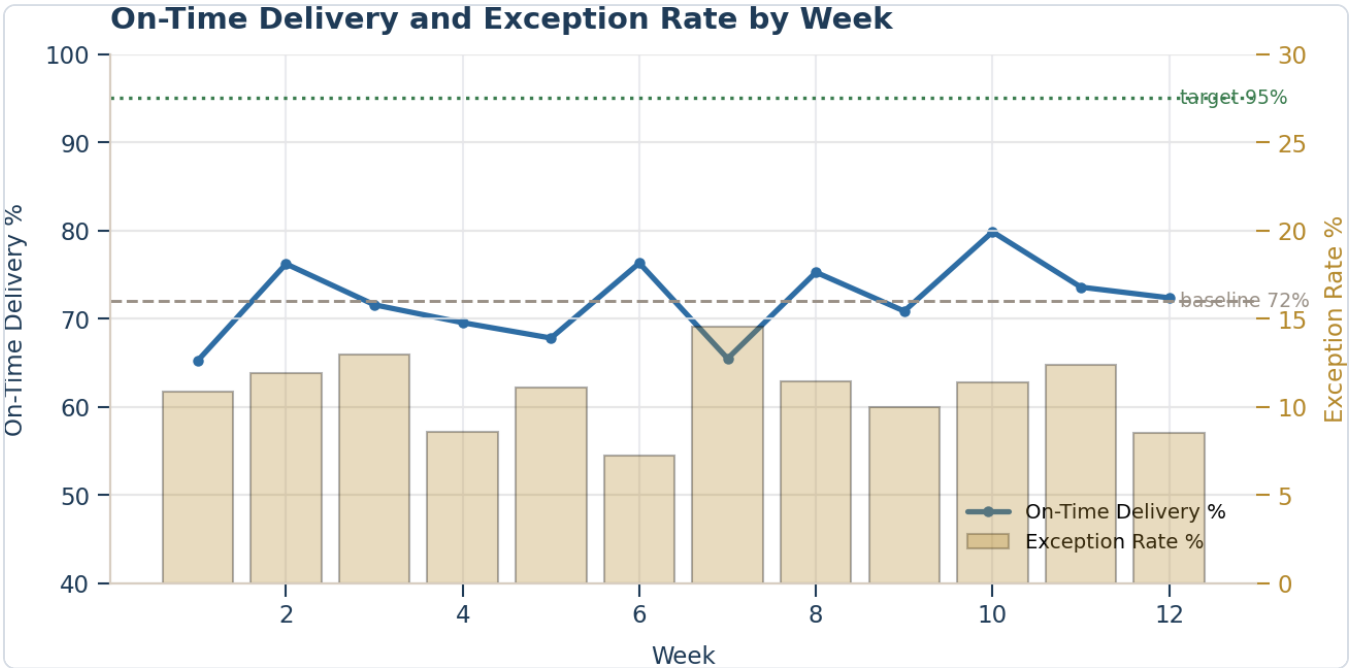


Figure 12.1 — Executive Overview, the weekly service and stability trend.

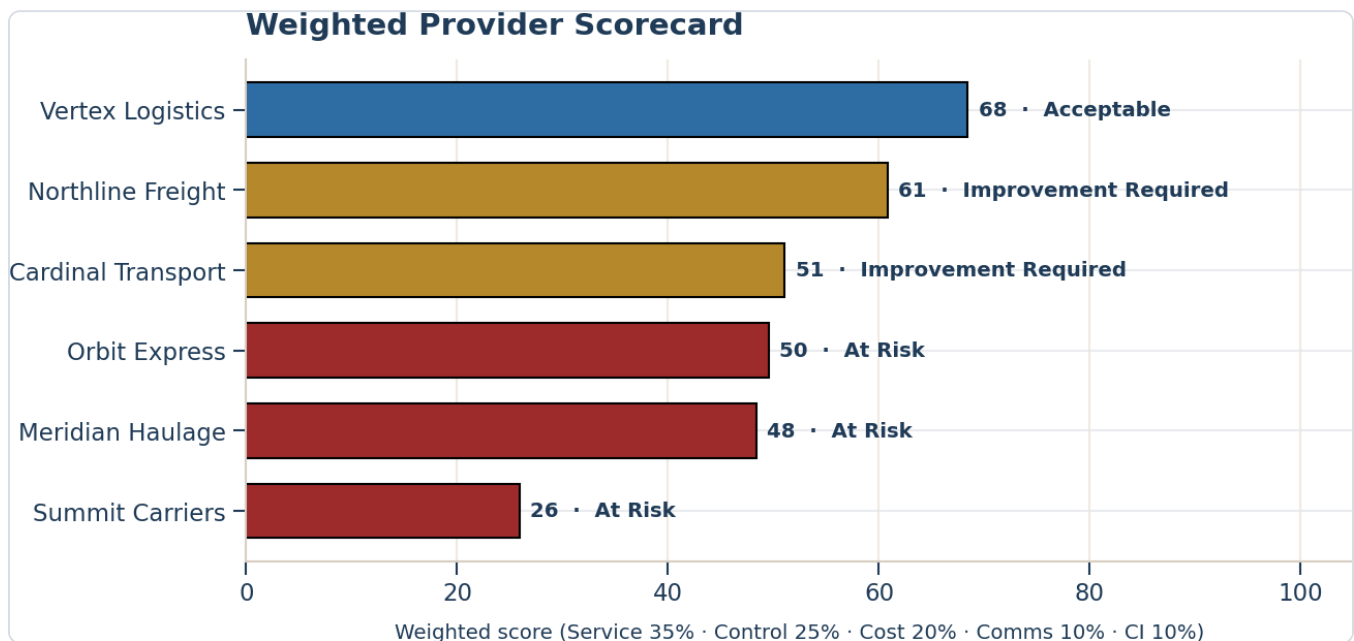


Figure 12.2 — Provider Performance, the weighted scorecard view used in monthly supplier reviews.

Design intent

A manager should be able to move from a KPI variance to the underlying provider, shipment, exception, owner and corrective action without leaving the reporting pack. Reporting that ends at the KPI creates awareness; reporting that ends at the owner creates improvement.

SECTION 13

Implementation Roadmap

An eight-week sequence that deliberately establishes definitions and data quality before deploying reporting.

Phase	Timing	Activities
Define	Week 1	Confirm scope, identify stakeholders, define KPIs and service standards, establish ownership and governance requirements
Clean	Weeks 2–3	Standardise shipment data, provider names, locations and service types; introduce reason codes; validate timestamps; identify documentation gaps
Control	Weeks 4–5	Launch exception workflow, establish ownership rules and escalation thresholds, introduce corrective-action tracking
Measure	Weeks 6–7	Deploy the KPI model, launch the provider scorecard, build the dashboard, establish the weekly operating review
Govern	Week 8	Launch monthly provider review, review recurring failure modes, track corrective actions, establish the improvement backlog
Improve	Ongoing	Refine thresholds, automate repeatable reporting, improve data quality, introduce predictive monitoring only after process discipline is established

Why measurement precedes automation

Automating an undefined process simply produces incorrect information faster. The roadmap deliberately places data standardisation and exception control ahead of reporting, and reporting ahead of prediction.

SECTION 14

Business Impact

Because this is a simulated case study, I make no claim of realised savings. What follows is the measurable value the operating model is designed to create, expressed so that it can be verified after implementation.

Dimension	Current state	Target state
KPI definitions	Six provider formats, no common baseline	One published KPI dictionary applied network-wide
Root-cause visibility	Late delivery assumed to be carrier failure	Every exception classified to one of four ownership categories
Accountability	No named owner or due date	Owner and due date on every material exception
Exception closure	5.4 days, 42 open at period end	3.0 days with managed backlog
Reporting effort	Manual consolidation from six sources	Single standardised dataset and dashboard
Supplier reviews	Anecdotal, incident-led	Evidence-led using a weighted scorecard
Cost-to-service	Cost and service reviewed separately	Reviewed together by provider, site and service type

The measurable prize

The analysis identified that 46% of exceptions are internally caused and that roughly half of all shipments breach the provider cut-off. These are addressable without renegotiating a single carrier contract, which makes them the fastest available route to closing the 23-point gap to target.

The model is intentionally designed so that benefits are measured after implementation rather than assumed in advance.

SECTION 15

Consultant Recommendations

Sequenced from control to capability. Each tier depends on the one before it.

Immediate — establish control

- Publish the logistics KPI dictionary and apply it to all six providers.
- Standardise the 14 logistics reason codes across every site and provider.
- Assign an owner and due date to every material exception, starting with the 42 currently open.
- Introduce cut-off compliance alerts and daily late-release reporting, targeting the 41% of exceptions caused by EXC-01 and EXC-03.
- Establish the weekly operational performance review.

Medium term — build accountability

- Introduce monthly provider scorecards using the weighted model.
- Open a formal corrective-action plan with Summit Carriers, focused on transit performance rather than collection.
- Review the Next Day and Express service proposition, where premium cost is not matched by reliability.
- Track recurring failure modes and require closure evidence for material exceptions.

Long term — automate, then predict

- Integrate shipment, provider, invoice and operational data into one source.
- Automate exception alerts and recurring performance reports.
- Introduce structured cost-to-service analysis into commercial reviews.
- Consider predictive exception monitoring once data quality and process discipline are proven.

SECTION 16

Lessons Learned

A short consultant reflection on what this engagement design demonstrates.

Logistics performance is a process problem as well as a transport problem

Improving delivery performance requires understanding the complete order-to-delivery process. In this dataset, the largest single driver sat upstream of the carrier entirely.

Not every late shipment is a carrier failure

With 46% of exceptions internally caused, the operating model must establish causality before assigning accountability. Without that discipline, supplier reviews become adversarial and the real drivers persist.

Scorecards need a management process

A supplier scorecard has limited value if nobody reviews the results or follows up corrective actions. The scorecard is the input to governance, not a substitute for it.

Visibility should lead to decisions

A dashboard without ownership and follow-through simply creates another reporting layer. Every view in this design terminates in a named owner.

Measurement should come before automation, and automation before prediction

Reliable definitions and clean operational data are prerequisites for effective automation. Predictive monitoring becomes useful only once the underlying process and data are stable.

The consultant takeaway

The most valuable output of this engagement is not the dashboard. It is the reclassification of the problem: from a carrier performance issue to an internal process control issue. That reframing is what makes the improvement achievable.

APPENDIX

Data Dictionary & Assumptions

Shipment record structure

Field	Purpose
Shipment ID	Unique shipment reference
Order Date	Date the customer requirement was created
Release Date	Date and time the shipment was released
Pickup Date/Time	Actual provider collection
Promised Delivery	Agreed delivery commitment
Delivery Date/Time	Actual delivery
Provider	Logistics service provider
Origin / Destination	Shipping and delivery location
Service Type	Logistics service category
Shipment Cost	Cost associated with the shipment
Exception Code	Standardised failure category, 14 codes
Root Cause Category	Internal, Provider, Customer/External, Data
Owner	Accountable person or team
Closure Status	Open, pending, verified or closed

Companion workbook

Worksheet	Contents
Executive Dashboard	Headline KPIs, weekly trend, provider comparison
Provider Scorecard	Weighted scoring model with performance banding
Exception Analysis	Root-cause mix, reason-code detail, ownership and closure
Site & Route	Origin performance, cut-off compliance, service type analysis
KPI Dictionary	Definitions, units, purpose, targets and baselines
Shipment Data	Full 1,248-record synthetic dataset

Assumptions

- All data in this case study is synthetic and Apex Distribution Ltd. is fictional.

- The dataset was generated to match the stated operating profile: 1,248 shipments, 8 sites, 6 providers, 12 weeks, 72% on-time delivery and an 11% exception rate.
- Exception distribution is weighted toward late shipments, consistent with operational reality.
- Scorecard weights are illustrative and would be validated with stakeholders.
- KPI targets are indicative and would be agreed with operations and commercial teams.
- Actual benefits would be measured after implementation rather than assumed.
- The scenario uses no confidential employer information.

Capabilities demonstrated

Supply Chain Management · Logistics Management · 3PL and Vendor Management · KPI Design · Performance Management · Exception Management · Process Improvement · Operational Governance · Supply Chain Analytics · Excel and BI Reporting · ERP and WMS Process Understanding